

Load-Aware A* Path Planning for Quadruped Robots

Day 17 of 30 · Week 2: Refinement & Integration

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1. Nature of Work

Day 17 finalised the Methodology section — the last writing task for Member 1 before Week 3's focus shifts entirely to testing and data collection. All six subsections were reviewed one final time, parameter settings were added to the Experimental Setup subsection, and the complete 3-4 page Methodology document was compiled and shared with Member 4 for integration into the full paper. Part 2 of this report contains the complete Methodology exactly as submitted.

The parameter settings addition was the main writing task today — the original draft stated what parameters were used, but did not justify the choices. A dedicated subsection was added explaining why each value was selected: why 10x10 and not larger, why 20 scenarios, why $\alpha=1.0$, and why $\text{body_mass}=10$ kg. This makes the paper stronger because reviewers often ask about parameter sensitivity and this pre-empts those questions.

2. Tasks Completed

- **Finalise Methodology**
All 6 subsections reviewed and signed off. Version bumped to v2.0. Shared with Member 4 via Google Drive for paper integration.
- **Add all sections complete**
Complete section inventory confirmed: A. Problem Formulation, B. Quadruped Model, C. Standard A*, D. Load-Aware A*, E. Implementation Details, F. Experimental Setup. All cross-references checked.
- **Include parameter settings**
New subsection III-F-3 added: Parameter Settings and Justification. Covers grid size, scenario count, payload range, alpha selection, body mass, and seed validation rationale.

3. NEW CONTENT — PARAMETER SETTINGS SUBSECTION

III-F-3. Parameter Settings and Justification Table III summarise the fixed parameter settings used across all experiments, with justifications for each choice.

Experimental Parameter Justification

This table outlines the key parameters used in the robot path planning experiment, providing the selected value and the rationale behind its choice.

Parameter	Value	Justification
Grid Dimensions (R x C)	10 x 10	This is a standard benchmark size for grid path planning. It is large enough to support diverse obstacle configurations yet small enough to allow rapid iteration across all 80 experiment runs (20 scenarios $\times$ 4 algorithms).
Number of Scenarios	20	Provides statistical validity for paired t-tests ($p < 0.05$). The scenarios cover 5 distinct obstacle density levels (from 20% to 45%), with 4 unique variations for each density, which include changes in payload position and mass.
Robot Body Mass (m_b)	10.0 kg	This mass is representative of a medium-sized quadruped robot (e.g., smaller research platforms are 5-15 kg; Boston Dynamics Spot is 32 kg). It is kept constant to specifically isolate the effect of varying the payload mass.
Payload Mass (m_p) Range	3.0 - 9.0 kg	This range represents 30% to 90% of the robot's body mass. Below 3 kg, the stability gradient (S(n)) is too shallow to cause path divergence at the chosen stability weight ($\alpha=1.0$) (as determined on Day 8). Above 9 kg, the robot's Centre of Mass (CoM) is too close to the support polygon edge, which is physically unsafe.

Stability Weight (alpha)	1.0	Empirically selected after a tuning experiment (Day 7). It is the minimum alpha value that consistently causes a significant path divergence when the payload is ≥ 4 kg, without creating excessive detours. This value was confirmed across Scenarios 1–10.
Cell Size	1.0 m	Used as a normalised unit. The physical results are linearly scalable and are applicable to any physical grid with cell dimensions in the range of 0.5–2.0 m.
Foot Half-Span (hs)	0.4 m	$0.4 \times \text{cell_size} = 0.4 \text{ m}$. This positions the robot's feet at 40% of the cell width from the center. This value ensures the support polygon remains within a single grid cell, which aligns with the quasi-static movement assumption (A4).
Minimum Stability Threshold (M(n))	$M(n) > 0.05 \text{ m}$	This 5cm minimum margin from the support polygon edge is the critical threshold. Operating below this margin puts the robot at significant risk of tipping.

5. OBSERVATIONS & NOTES

Parameter justification is often what separates a strong paper from a weak one. Many papers report results but never explain why specific values were chosen. Adding the justification table forces clarity — it turned out that several choices ($hs = 0.4$, $\text{threshold} = 0.05 \text{ m}$) were implicit in the code but had never been explicitly stated in the paper. They are now. The Methodology is complete and consistent with the code. Every equation in the section corresponds to a line of code in the notebook. Every parameter value in Table III is exactly what is used in the experiments. This alignment between paper and implementation is what makes results reproducible and credible to reviewers